

1. Write a unittest class TestMath with one test method that checks if 2 + 3 equals 5.

```
import unittest
```

```
class TestMath(unittest.TestCase):  
    def test_addition(self):  
        self.assertEqual(2 + 3, 5)
```

2. Create a test class that uses setUp() to initialize a list [1, 2, 3] and tearDown() to print "Test completed". Verify that the list length is 3 inside your test method.

```
import unittest
```

```
class TestList(unittest.TestCase):  
    def setUp(self):  
        self.data = [1, 2, 3]  
  
    def tearDown(self):  
        print("Test completed")  
  
    def test_list_length(self):  
        self.assertEqual(len(self.data), 3)
```

3. Write a test class TestStringMethods with methods to test:  
"hello".upper() equals "HELLO"  
"hello".isupper() returns False

```
import unittest
```

```
class TestStringMethods(unittest.TestCase):  
    def test_upper(self):  
        self.assertEqual("hello".upper(), "HELLO")
```

```
def test_isupper(self):  
    self.assertFalse("hello".isupper())
```

4. Use `assertRaises` to verify that dividing by zero (`10 / 0`) raises a `ZeroDivisionError`.

```
import unittest
```

```
class TestException(unittest.TestCase):  
    def test_zero_division(self):  
        with self.assertRaises(ZeroDivisionError):  
            x = 10 / 0
```

5. Create two test classes (`TestAdd` and `TestSubtract`) and combine them into a single test suite using `unittest.TestSuite`. Run the suite using `unittest.TextTestRunner()`.

```
import unittest
```

```
class TestAdd(unittest.TestCase):  
    def test_add(self):  
        self.assertEqual(5 + 3, 8)
```

```
class TestSubtract(unittest.TestCase):  
    def test_subtract(self):  
        self.assertEqual(10 - 4, 6)
```

```
if __name__ == "__main__":  
    suite = unittest.TestSuite()  
  
    suite.addTest(unittest.makeSuite(TestAdd))  
    suite.addTest(unittest.makeSuite(TestSubtract))  
  
    runner = unittest.TextTestRunner()  
    runner.run(suite)
```